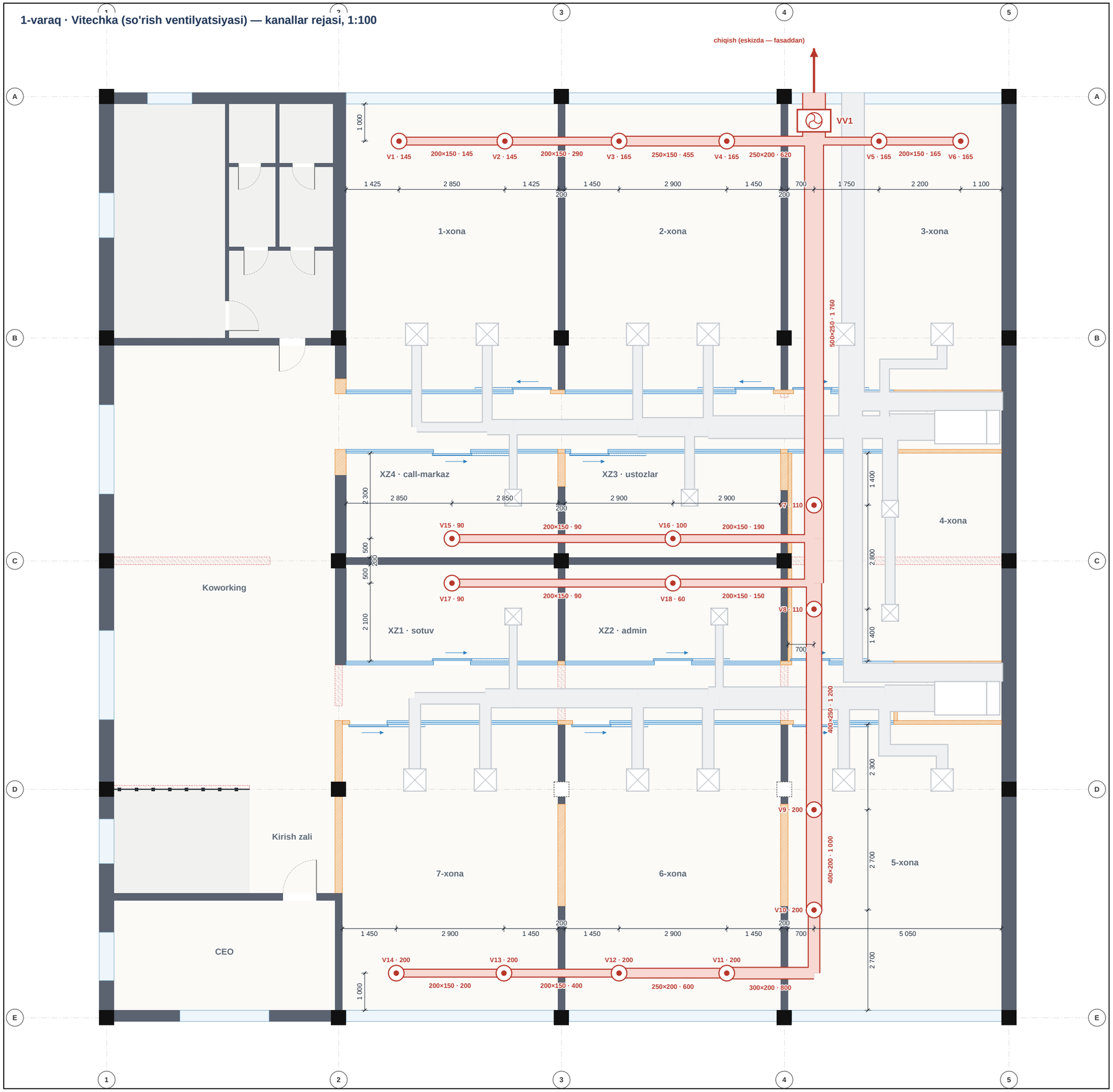
## 1-varaq · Vitechka (so'rish ventilyatsiyasi) — kanallar rejasi, 1:100




## VITECHKA — SO'RISH KANALLARI (buyurtmachi eskizi bo'yicha)

so'rish kanali (kesim × havo, m³/soat) — so'rish panjarasi (klapan) V…
VV1 — kanalli ventilyator — konditsioner kanallari (2-varaq, xira)

### Panjaralar — 18 ta, jami 2 710 m³/soat

| Kod | Xona | x, mm | y, mm | m³/soat | Ø, mm |
|---|---|---|---|---|---|
| **V1** | 1-xona | 7 675 | 1 000 | 145 | 200 |
| **V2** | 1-xona | 10 525 | 1 000 | 145 | 200 |
| **V3** | 2-xona | 13 600 | 1 000 | 165 | 200 |
| **V4** | 2-xona | 16 500 | 1 000 | 165 | 200 |
| **V5** | 3-xona | 20 600 | 1 000 | 165 | 200 |
| **V6** | 3-xona | 22 800 | 1 000 | 165 | 200 |
| **V7** | 4-xona | 18 850 | 10 800 | 110 | 200 |
| **V8** | 4-xona | 18 850 | 13 600 | 110 | 200 |
| **V9** | 5-xona | 18 850 | 19 000 | 200 | 200 |
| **V10** | 5-xona | 18 850 | 21 700 | 200 | 200 |
| **V11** | 6-xona | 16 500 | 23 400 | 200 | 200 |
| **V12** | 6-xona | 13 600 | 23 400 | 200 | 200 |
| **V13** | 7-xona | 10 500 | 23 400 | 200 | 200 |
| **V14** | 7-xona | 7 600 | 23 400 | 200 | 200 |
| **V15** | XZ4 · call-markaz | 9 100 | 11 700 | 90 | 200 |
| **V16** | XZ3 · ustozlar | 15 050 | 11 700 | 100 | 200 |
| **V17** | XZ1 · sotuv | 9 100 | 12 900 | 90 | 200 |
| **V18** | XZ2 · admin | 15 050 | 12 900 | 60 | 200 |

x — chap (g'arbiy) fasad ichki yuzidan, y — yuqori fasad ichki yuzidan (1-varaqdagi o'qlar bo'yicha). So'rish havosi — xonaga konditsioner orqali kelayotgan tashqi havoning (30%) 90% i: xonalar biroz ortiqcha bosimda, koridordan chang kirmaydi.

### Kanallar (galvanizlangan po'lat, to'rtburchak, tekis yotqiziladi)

| Liniya | m³/soat | Kesim, mm | Uzunlik, m |
|---|---|---|---|
| 1–2-xona, fasad bo'ylab | 145–290 | **200 × 150** | 5.9 |
| 1–2-xona, fasad bo'ylab | 455 | **250 × 150** | 2.9 |
| 1–2-xona, fasad bo'ylab | 620 | **250 × 200** | 2.4 |
| 3-xona, fasad bo'ylab | 165–330 | **200 × 150** | 4.0 |
| 7–6-xona, fasad bo'ylab | 200–400 | **200 × 150** | 6.0 |
| 7–6-xona, fasad bo'ylab | 600 | **250 × 200** | 2.9 |
| 7–6-xona, fasad bo'ylab | 800 | **300 × 200** | 2.4 |
| XZ4, XZ3 (o'rta devor yonida) | 90–190 | **200 × 150** | 9.8 |
| XZ1, XZ2 (o'rta devor yonida) | 90–150 | **200 × 150** | 9.8 |
| magistral (koridor oxirlari) | 800 | **300 × 200** | 1.7 |
| magistral (koridor oxirlari) | 1 000 | **400 × 200** | 2.7 |
| magistral (koridor oxirlari) | 1 200–1 310 | **400 × 250** | 6.1 |
| magistral (koridor oxirlari) | 1 460–1 760 | **500 × 250** | 11.9 |
| magistral (koridor oxirlari) | 2 710 | **600 × 300** | 0.3 |

**VV1:** kanalli ventilyator 3 000 m³/soat, ~400 Pa (to'rtburchak, 600×300 ga mos), shovqin so'ndirgich bilan; kanallar jami ~69 m.

### O'rnatish

- Kanallar shift ostida, osmalarda (har 2–2.5 m); pastki qirrasi ≥ 2.9 m. Shisha devorlar ustidagi GKL (2.5–3.5 m) va mavjud devorlardan teshik orqali o'tadi; teshik atrofi germetiklanadi.
- Ventilyator osma rezina amortizatorlarda, kanalga egiluvchan ulanadi; har liniya boshida rostlash klapani, panjaralar sozlanadigan (havo — jadval bo'yicha).
- Kanallar 2-varaqdagi konditsioner kanallari bilan 5 joyda kesishadi — so'rish kanali yuqorida, berish kanali ostida (2-varaq, ①–⑤).

**Diqqat — chiqish joyi:** eskizda VV1 chiqishi va konditsionerning tashqi havo olish panjarasi 3-xona fasadida, oralig'i ~1 050 mm — chiqarilgan havo qaytib so'rib olinadi. Me'yor: olish va chiqarish orasi ≥ 8 m. Fasad o'zgarmasligi kerak (oyna ustida faqat ~400 mm devor bor). **Tavsiya:** tashqi havo buyurtmachi qarori bilan 3-xona ko'cha tomonidan qoladi — shuning uchun VV1 chiqishini to'g'ridan tomga (tom sathidan ≥ 1 m baland, soyabon bilan) chiqarish kerak.

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|---|---|
| Chizma | **Vitechka — so'rish kanallari rejasi** |

Ish chizmasi — ustalar uchun. Buyurtmachi eskizi (15.10) va qarorlari (16.10) asosida. Toshkent sh., Xadra — 2-qavat

| Masshtab | Sana | Versiya | Varaq |
|---|---|---|---|
| 1:100 | 16.10.2026 | v1.2 | 1 / 2 |

Bajardi: _____ Tekshirdi: _____

## 2-varaq · Konditsioner (kanalli) va tashqi havo — kanallar rejasi, 1:100




## KANALLI KONDITSIONER — BERISH KANALLARI (16.10)

- sovuq havo kanali (kesim × m³/soat)
- tashqi havo kanali (30%, blokka)
- vitechka (1-varaq, xira)
- shift diffuzori D… (4 tomonga)
- KK — ichki blok, kladovka shifti ustida
- ①–⑤ kesishuvlar

### Ichki bloklar — kladovka (KL1, KL2) shifti ustida, devorga taqalgan (96 000 BTU deb olindi)

| Blok | Xonalar | Havo | Tashqi havo |
|---|---|---|---|
| **KK1** | 1, 2, 3, 4, XZ3 · ustozlar, XZ4 · call-markaz | 5 000 | 1 500 (30%) |
| **KK2** | 5, 6, 7, XZ1 · sotuv, XZ2 · admin | 5 000 | 1 500 (30%) |

### Quvvat tekshiruvi (kW, eng issiq payt)

| Blok | Xonalar | + tashqi havo 30% | Jami | 96k (24.4 kW issiqda) |
|---|---|---|---|---|
| **KK1** | 24.5–30.3 | 7.0 | **37.4** | yetmaydi (−12.9 kW) |
| **KK2** | 24.2–29.2 | 7.0 | **36.3** | yetmaydi (−11.8 kW) |

### Xonalar: havo, tashqi havo, so'rish, $CO_2$

| Xona | Odam | Havo | Tashqi (30%) | So'rish | $CO_2$, ppm |
|---|---|---|---|---|---|
| 1-xona (KK1) | 25 | 1 060 | 318 | 290 | 1 840 |
| 2-xona (KK1) | 25 | 1 220 | 366 | 330 | 1 650 |
| 3-xona (KK1) | 25 | 1 220 | 366 | 330 | 1 650 |
| 4-xona (KK1) | 19 | 800 | 240 | 220 | 1 840 |
| XZ4 · call-markaz (KK1) | 6 | 320 | 96 | 90 | 1 550 |
| XZ3 · ustozlar (KK1) | 9 | 380 | 114 | 100 | 1 840 |
| 5-xona (KK2) | 25 | 1 480 | 444 | 400 | 1 430 |
| 6-xona (KK2) | 25 | 1 470 | 441 | 400 | 1 440 |
| 7-xona (KK2) | 25 | 1 470 | 441 | 400 | 1 440 |
| XZ1 · sotuv (KK2) | 6 | 340 | 102 | 90 | 1 480 |
| XZ2 · admin (KK2) | 4 | 230 | 69 | 60 | 1 460 |

### Diffuzorlar — 18 ta

| Kod | Xona | x | y | m³/soat | Panel | Patrubok |
|---|---|---|---|---|---|---|
| **D1** | 1-xona | 8 150 | 6 200 | 530 | 600×600 | 250×200 |
| **D2** | 1-xona | 10 050 | 6 200 | 530 | 600×600 | 250×200 |
| **D3** | 2-xona | 14 100 | 6 200 | 610 | 600×600 | 250×200 |
| **D4** | 2-xona | 16 000 | 6 200 | 610 | 600×600 | 250×200 |
| **D5** | 3-xona | 19 650 | 6 200 | 610 | 600×600 | 250×200 |
| **D6** | 3-xona | 22 300 | 6 200 | 610 | 600×600 | 250×200 |
| **D7** | 4-xona | 20 900 | 10 900 | 400 | 450×450 | 400×200 |
| **D8** | 4-xona | 20 900 | 13 700 | 400 | 450×450 | 250×150 |
| **D9** | 5-xona | 19 650 | 18 200 | 740 | 600×600 | 300×200 |
| **D10** | 5-xona | 22 300 | 18 200 | 740 | 600×600 | 300×200 |
| **D11** | 6-xona | 14 100 | 18 200 | 735 | 600×600 | 300×200 |
| **D12** | 6-xona | 16 000 | 18 200 | 735 | 600×600 | 300×200 |
| **D13** | 7-xona | 8 100 | 18 200 | 735 | 600×600 | 300×200 |
| **D14** | 7-xona | 10 000 | 18 200 | 735 | 600×600 | 300×200 |
| **D15** | XZ4 · call-markaz | 10 750 | 10 600 | 320 | 450×450 | 200×150 |
| **D16** | XZ3 · ustozlar | 15 500 | 10 600 | 380 | 450×450 | 250×150 |
| **D17** | XZ1 · sotuv | 10 750 | 13 800 | 340 | 450×450 | 200×150 |
| **D18** | XZ2 · admin | 16 300 | 13 800 | 230 | 450×450 | 200×150 |

| Magistral | m³/soat | Kesim, mm | Uzunlik, m |
|---|---|---|---|
| K1 koridori (KK1 dan) | 530 | **250 × 150** | 1.9 |
| K1 koridori (KK1 dan) | 1 060 | **400 × 200** | 0.7 |
| K1 koridori (KK1 dan) | 1 380 | **400 × 250** | 3.4 |
| K1 koridori (KK1 dan) | 1 990–2 370 | **500 × 300** | 1.9 |
| K1 koridori (KK1 dan) | 2 980 | **600 × 300** | 3.6 |
| K1 koridori (KK1 dan) | 3 590 | **600 × 350** | 1.1 |
| K1 koridori (KK1 dan) | 4 200 | **600 × 400** | 0.1 |
| K1 koridori (KK1 dan) | 5 000 | **700 × 400** | 1.2 |
| K2 koridori (KK2 dan) | 735 | **300 × 200** | 1.9 |
| K2 koridori (KK2 dan) | 1 470 | **500 × 250** | 0.8 |
| K2 koridori (KK2 dan) | 1 810–2 545 | **500 × 300** | 5.3 |
| K2 koridori (KK2 dan) | 3 280–3 510 | **600 × 350** | 3.6 |
| K2 koridori (KK2 dan) | 4 250 | **600 × 400** | 1.1 |
| K2 koridori (KK2 dan) | 4 990 | **700 × 400** | 1.4 |
| tashqi havo → KK1 | 1 500 | **500 × 250** | 4.0 |
| tashqi havo: 3-xona fasadi → KK2 | 1 500 | **500 × 250** | 11.3 |
| tashqi havo: 3-xona fasadi → KK2 | 3 000 | **600 × 300** | 8.3 |

Kanallar jami: berish ~72 m (patrubokli), tashqi havo ~24 m, vitechka ~69 m — hammasi ~165 m (narx: 300 000 so'm/m). Qaytish havosi (QP): koridor shiftidagi panjara 600 × 300, kladovka devori yonida (logoli variantda reykalar ortida) — shift ustidagi bo'shliq orqali blokning qaytish kamerasiga.

**Diqqat:** (1) 2 × 96k yetmaydi — har blokka ~37 kW (~130k BTU) kerak; 96k issiqda ~24.4 kW beradi. (2) 30% tashqi havoda to'la sinfda $CO_2$ ~1 450–1 850 ppm (me'yor ≤ 1 000–1 200): darsdan keyin shamollatish yoki tashqi havoni ~45–50% ga oshirish (blok shunga mos tanlansa) tavsiya.

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|---|---|
| Chizma | **Konditsioner (kanalli) va tashqi havo** |

Ish chizmasi — ustalar uchun. Buyurtmachi eskizi (15.10) va qarorlari (16.10) asosida. Toshkent sh., Xadra — 2-qavat

| Masshtab | Sana | Versiya | Varaq |
|---|---|---|---|
| 1:100 | 16.10.2026 | v1.2 | 2 / 2 |

Bajardi: _____ Tekshirdi: _____